Sucrose Intake and Preference in Rover and Sitter *Drosophila melanogaster* Males Measured with the CAFE Assay

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Introduction

Food was critical to the survival of any organism. In *Drosophila melanogaster*, commonly down as the fruit fly, searched for food depending on the different genetic mutations it had. Rovers were a mutation where the fly would continue to move around regardless of access to food; whereas sitters (*for*-S) found one food source and stayed there (Sokolowski, 1980). This was because of how the *for* gene affected the production of the protein Kinase (de Belle et al., 1989). Rovers had higher PKG activity than sitters which changed how neurons in the fly's brain fired, in turn changing the behavior of the fly (Osborne et al., 1997; Sokolowski, 2001).

Interestingly, this mutation did not just effect the movement of the fly; in fact, mutations to the *for* gene changed how the fly responded to different concentrations of (Scheiner et al., 2004). How much they ate was different too, for example: larvae ate less than sitter larvae when food was plentiful (Kaun et al., 2007; Allen et al., 2017).

In this experiment the scientist wanted to quantify the difference in food intake depending on the gene of the fly. The scientist measured the intake of food in adult flies with the Capillary Feeder (CAFE) assay, where flies drank colored liquid out of top of the vial (Ja et al., 2007). As the flies ate the volume in the tube decreased and the scientist was able to calculate the difference in volume. In this experiment, vials of rover (*for*-R) and sitter (*for*-S) males were given 15% sucrose and a control of plain water, and each food was offered in both red and blue dye so that a preference for a color could be told apart from a preference for sucrose. It was hypothesized that both genotypes would drink more 15% sucrose than water, and that rover males would drink more sucrose than sitter males.

Materials and Methods

Materials

Male *D. melanogaster* from the rover (*for*-R) and sitter (*for*-S) strains were used. Food was offered in 5 µL glass tubes which were also the tool used to measure how much the flies had eaten. A 15% sucrose solution and a control of water with no sucrose were each dyed red or blue with food. A grey and black sharpie marker was used to mark the starting levels of liquid and the silver marker was used to put a band near the top of each sucrose tube. Flies were anesthetized with CO_2 on a fly pad and sorted with a paintbrush. At the end of the assay Dr. Krill photographed the tubes next to a stainless-steel ruler with a smartphone camera.

Animal Preparation

The scientist anesthetized each of the flies using CO_2 and then sorted them by sex, this was done to ensure cross sex appetite changes were not influencing results. Ten males were put into each vial, and each vial was labeled with genotype and sex (Figure 1C).

CAFE Assay

The tubes were filled by dipping the lower end into the appropriate solution with food coloring (Figure 1A, 1B), and the starting food level was marked in black sharpie. All four tubes were pushed through the cotton plug of each vial with the open tips facing down where the flies could reach them: a red control, a blue control, a red 15% sucrose, and a blue 15% sucrose tube. The flies fed for 24 h at room temperature. Finally, the tubes were taken out, laid side by side and photographed from above (Figure 2).

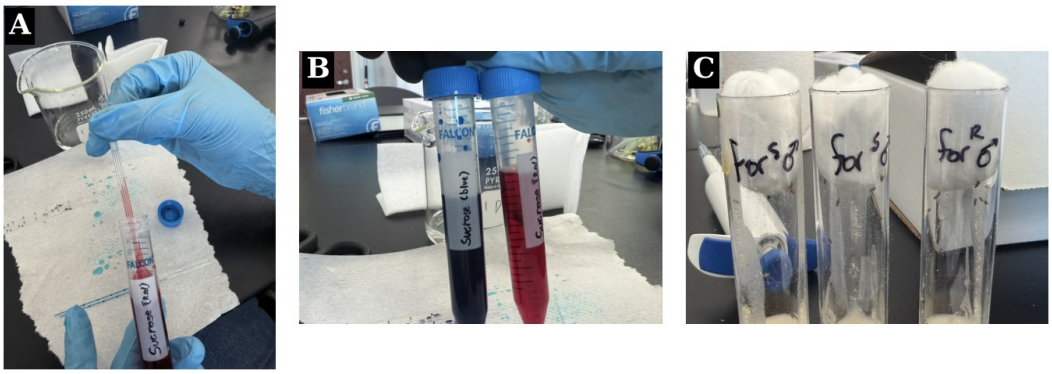


Figure 1: Set up of the CAFE assay. Panel (A) showed a glass tube being filled with red 15% sucrose, with the marked blue tubes on the paper towel, Panel (B) showed the blue and red dyed sucrose tubes used to fill the glass tubes, Panel (C) showed the sexed male flies labeled by genotype (*for*-S and *for*-R) and sealed with cotton plugs.

Measuring Food Consumption

Food consumption was measured based on the photograph in Figure 2. For each tube, the distance in pixels between the t = 0-hour black mark and the end of the dyed liquid when photographed after 24 hours. The Pixels were converted to millimeters using the ruler in the photo as a reference. The photo was taken at a slight angle, so each distance was also scaled by the pixel length of its own tube. Volume was then calculated as volume = distance × 5 µL / 127 mm, or 0.039 µL per mm, and intake per fly was estimated by dividing by ten flies.

Data and Results

Table 1: Raw CAFE data for each tube by genotype, food and dye color. Each distance was how far the food level dropped over the 24 h trial, from the t = 0 h black mark to the end of the dyed liquid, converted into millimeters using the ruler in the photo as a reference. When a tube had two black marks, the bottom mark that was closer to the food was used. Volume was calculated using 5 µL per 127 mm tube, and volume per fly assumed ten males per jar. Tube numbers matched Figure 2.

#	Genotype	Food	Dye	Distance (mm)	Volume (µL)	µL / fly
1	*for*-R	Control (water)	Blue	29.2	1.15	0.115
2	*for*-R	Control (water)	Red	25.7	1.01	0.101
3	*for*-R	15% sucrose	Red	28.3	1.12	0.112
4	*for*-R	15% sucrose	Blue	51.9	2.05	0.205
5	*for*-R	Control (water)	Blue	36.0	1.42	0.142
6	*for*-R	15% sucrose	Red	33.8	1.33	0.133

#	Genotype	Food	Dye	Distance (mm)	Volume (µL)	µL / fly
7	*for*-R	Control (water)	Red	30.7	1.21	0.121
8	*for*-R	15% sucrose	Blue	52.7	2.08	0.208
9	*for*-S	Control (water)	Red	22.2	0.87	0.087
10	*for*-S	Control (water)	Blue	21.6	0.85	0.085
11	*for*-S	15% sucrose	Red	22.0	0.87	0.087
12	*for*-S	15% sucrose	Blue	42.6	1.68	0.168
13	*for*-S	15% sucrose	Red	46.4	1.82	0.182
14	*for*-S	15% sucrose	Blue	59.7	2.35	0.235
15	*for*-S	Control (water)	Blue	29.3	1.15	0.115
16	*for*-S	Control (water)	Red	38.9	1.53	0.153

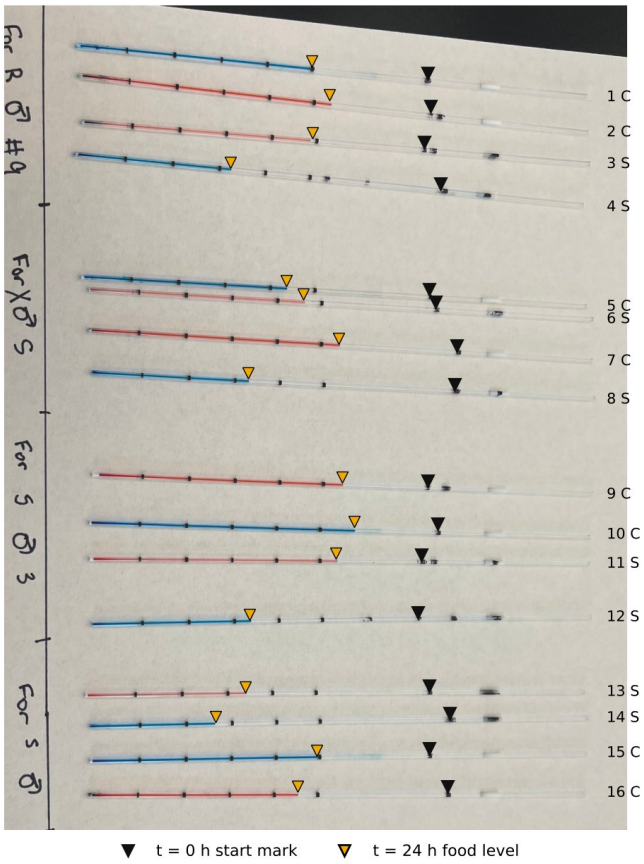


Figure 2: This photo showed the tubes after 24 h of feeding, with the rover on top and the sitter tubes below. The grey marking indicated that the liquid inside was the control. The black triangles showed the starting food level at t = 0 h and the yellow triangles showed the end of the dyed liquid at 24 hours.

Table 2: This table showed the mean from the four tubes in the same jar, two dyed red and two dyed blue, and the standard deviation and standard error was listed for each group. The individual measurements behind every mean were listed in Table 1.

Genotype	Food	n (tubes)	Mean (µL)	SD (µL)	SEM (µL)
for-R	15% sucrose	4	1.64	0.49	0.25

Genotype	Food	n (tubes)	Mean (µL)	SD (µL)	SEM (µL)
for-R	Control (water)	4	1.20	0.17	0.08
for-S	15% sucrose	4	1.68	0.61	0.31
for-S	Control (water)	4	1.10	0.32	0.16

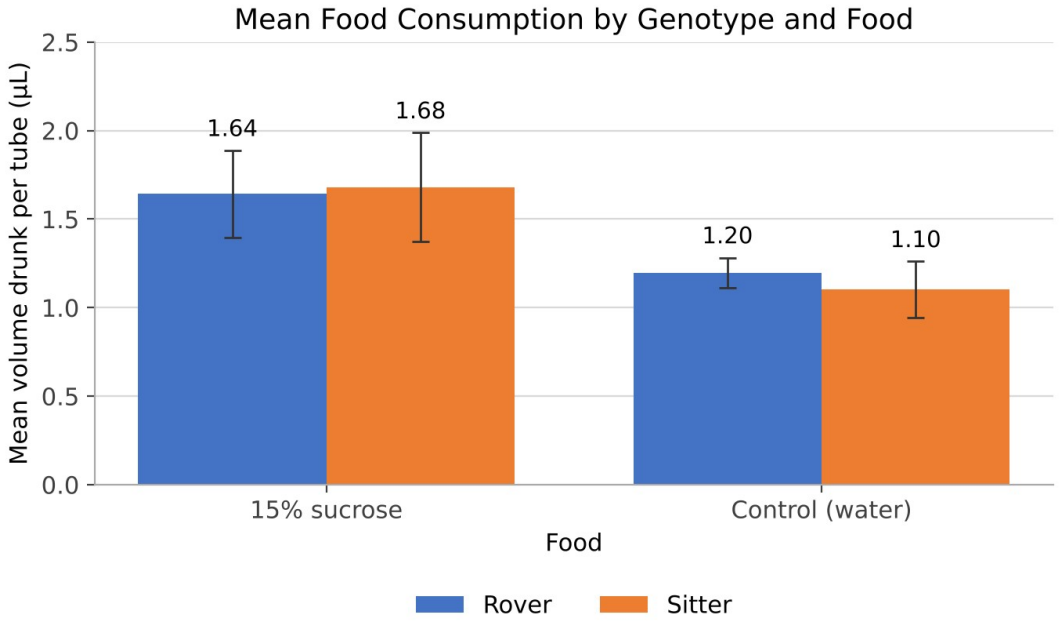


Figure 3: This graph showed the food consumption of the rover and sitter males over the 24 hour period. The blue bars showed the rover males and orange bars showed sitter males. Each bar represented the mean volume drunk per tube of either 15% sucrose or the water control. Both rovers and sitters drank more 15% sucrose than water, and the two genotypes drank about the same amount.

Written Results

All 16 tubes dropped by a lot in the 24 h period (Figure 2). The change in food in terms of milimeters ranged from 21.6 mm to 59.7 mm. The total intake per jar across all eight tubes was 11.36 µL for Rovers and 11.13 µL for Sitters.

As expected, both genotypes drank more 15% sucrose than water, supporting the scientist's hypothesis (Figure 3; Table 2). The rover males drank 1.64 ± 0.25 µL of sucrose and 1.20 ± 0.08 µL of water per tube, while sitter males drank 1.68 ± 0.31 µL of sucrose and 1.10 ± 0.16 µL of water.

Red sucrose (1.28 ± 0.20 µL) and blue sucrose (2.04 ± 0.14 µL) were not consumed equally, as the blue sucrose tubes lost more volume. The blue control tubes lost 1.14 ± 0.12 µL, while red control tubes lost 1.16 ± 0.14 µL, about the same as each other. In each group of four tubes, the blue sucrose was the tube that lost the most volume (Table 1).

Discussion

The point of this experiment was to determine whether male sitters or rovers had higher food consumption rates.

Both rovers and sitters drank more 15% sucrose than water; however, the two genotypes drank about the same amount (Table 2; Figure 3). This could have meant that flies preferred sugar, and went against the previous assumption that rovers would drink more sucrose than sitters (Scheiner et al., 2004). These results could also have gone against previous literature because of the small sample size which was not necessarily representative of the entire population. The scientist also did not account for evaporation in this experiment.

Interestingly, a majority of the sucrose preference was from the blue sucrose, while red and blue water were about equal. It was hypothesized that this could have been a real color preference or simply a problem with the blue sucrose tubes.

Because the *for* gene changed behavior through PKG which changed how excitable neurons (Osborne et al., 1997), and PKG signaling was shared across animals (Ben-Shahar et al., 2002), it could help researchers understand how genes influence appetite and food choice in other species, including humans.

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